AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 2 · STUDENT LESSON

Dot products and weighted sums

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Under schema [width, height], why must [4, 2] and [2, 4] remain distinct?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Calculate a dot product by matching feature positions. Try the recall before opening notes.

Your learning target

I can calculate a **dot product** by matching feature positions.

I can explain how a **weight** changes a **score** with other **input**s fixed.

I can distinguish a **weight**ed **score** from a probability or trained model.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Calculate a **dot product** by matching feature positions

Explain how a **weight** changes a **score** with other **input**s fixed

Distinguish a **weight**ed **score** from a probability or trained model

Retrieve what you already know

> Under schema [width, height], why must [4, 2] and [2, 4] remain distinct?

My starting explanation

This week's end goal

By the end, I can explain **dot product**s and **weight**ed sums using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Dot products and weighted sums case. Separate observations, given facts and assumptions before choosing an explanation.

A small model can combine several measurements into one **score**. Let x = [4, 2] contain fictional width and height measurements in cm. Set **weight**s w = [2, -1].

Multiply matching positions, then add: s = (2 times 4) + (-1 times 2) = 6. The letter x names the **input** vector.

We call this calculation a **dot product**, written w dot x. The vectors must have the same number of components. Each **weight** belongs to one named feature.

These hand-selected **weight**s demonstrate a computation; no **training** has happened. The result 6 is a **score**, not a probability or a claim that this parcel is safe.

Feature	Weight times input	Contribution
Width	2 times 4	8
Height	-1 times 2	-2
Total	8 plus -2	6

The coefficients carry **score**-per-cm units in this example, making both **contribution**s compatible **score** units. A **weight**ed sum needs a consistent definition of units.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Dot products and weighted sums. Connect each labelled input, deciding step and result.

Keep w = [2, -1] fixed. For A = [4, 2], the **score** is 6. For B = [2, 4], it is 4 - 4 = 0. For C = [3, 3], it is 6 - 3 = 3. Swapping **input** positions without swapping their matching **weight**s changes the computation.

A zero **score** can come from cancellation, even when neither **input** is zero.

```
def dot(weights, inputs):
    if len(weights) != len(inputs):
        raise ValueError("Mismatched dimensions")
    return sum(w * x for w, x in zip(weights, inputs))

for row in [[4, 2], [2, 4], [3, 3]]:
    print(dot([2, -1], row))  # 6, 0, 3
```

The length check matters: Python zip alone stops at the shorter list and could silently omit a feature. The function assumes numeric **input**s already validated under the schema.

First predict the three printed values on paper, then compare.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Dot products and weighted sums. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Increase only the right **input** from 5 to6.

Predicted result

Prediction: The **score** becomes 6-6=0.

Why the result follows

Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Dot products and weighted sums to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Weight	A numerical coefficient multiplying one input component

Word	Meaning
Dot product	The sum of products of corresponding vector components
Contribution	The product of one input component and its matching weight
Score	A numerical output whose interpretation must be specified

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Dot products and weighted sums, predict a step, then compare with its explanation.

Multiply matching components then add

POSITION	CONTRIBUTION RULE	WORKED CONTRIBUTION
1 width	2 times width	2 times 4 gives 8
2 height	-1 times height	-1 times 2 gives -2
Add both	8 plus -2	Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Keep w = [2, -1]. Change [4, 2] to [4, 3]. Compute both contributions and explain the score change.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Dot products and weighted sums

Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

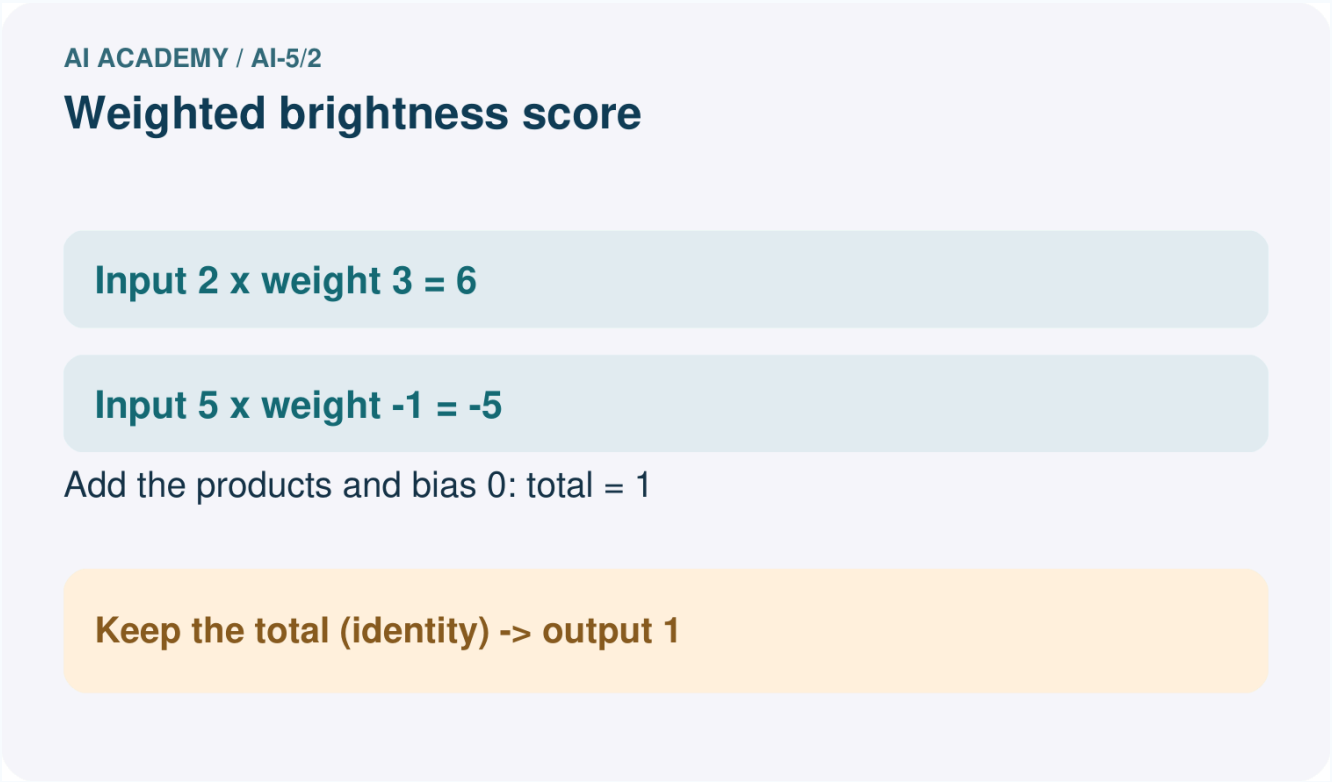
The facts we will use

Supplied facts

Inputs [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/2

Dot products and weighted sums

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply first input

Left contribution=2 x 3=6.



2. Multiply second input

Right contribution=5 x (-1)=-5.



3. Add contributions

Weighted sum=6-5=1; this is a score, not a probability.

CHANGE ONE THING

Increase only the right input from 5 to6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Multiply first input	Left contribution=2×3=6.
2. Multiply second input	Right contribution=5×(-1)=-5.
3. Add contributions	Weighted sum=6-5=1; this is a score, not a probability.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **weight** controls how strongly an **input** contributes to the **score**. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

What this example does not establish

Keep the stated limit

These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

Keep the stated limit

Read the labelled picture: Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Increase only the right **input** from 5 to6. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Dot products and weighted sums claim and explain a repair using the supplied evidence.

Holding other **input**s fixed, raising width by 1 cm raises this **score** by 2; raising height by 1 cm lowers it by 1. A negative **weight** reverses the direction of that feature **contribution**.

It does not label the feature as bad, and its sign alone does not explain cause in the world.

Check the claim against the facts

Unit changes affect coefficients. If height is written in mm instead of cm while width remains in cm, use height **weight** -0.1 instead of -1 to preserve the same **score**s.

For [4 cm, 20 mm], 2 times 4 plus -0.1 times 20 still gives 6. Raw coefficient magnitudes cannot establish feature importance when units differ.

Locate the earliest unsupported step

Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised **weight**s and **score**.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Dot products and weighted sums with support, then explain the deciding step in your own words.

For w = [2, -1] and x = [3, 5], write both **contribution**s and their sum.

Keep w = [2, -1]. Change [4, 2] to [4, 3]. Compute both contributions and explain the score change.

Explain your prediction to a partner before comparing results.

Practise with less help: Explain cancellation

Use workbook W2 for the connected example “**Dot product**s and **weight**ed sums”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Dot product**s and **weight**ed sums. Proposed explanation: “Multiply the sums instead”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Run the investigation

> Investigate Dot products and weighted sums. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

```
def dot(weights, inputs):
    if len(weights) != len(inputs):
        raise ValueError("Mismatched dimensions")
    return sum(w * x for w, x in zip(weights, inputs))

for row in [[4, 2], [2, 4], [3, 3]]:
    print(dot([2, -1], row))  # 6, 0, 3
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Dot products and weighted sums. Explain why each step is justified.

Explain → apply → challenge

For w = [2, -1] and x = [3, 5], write both **contribution**s and their sum.

Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised **weight**s and **score**.

> Verify that [4, 2] and [5, 4] both **score** 6 with w = [2, -1]. Can a later step receiving only this **score** distinguish those two records?

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **dot product**s and **weight**ed sums **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a **dot product** by matching feature positions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

A **weight**ed sum can combine **evidence**, but it discards information: [4, 2] and [5, 4] both produce **score** 6 under these **weight**s. A downstream rule receiving only that **score** cannot recover which of these vectors was used.

To classify, we would also need a stated decision rule and labelled evaluation examples.

A teammate calls score 6 a 60 percent chance of class 1. What is missing from that claim?

Next week adds an offset called a bias and states exactly how a **score** becomes a class **prediction**.

Verify that [4, 2] and [5, 4] both **score** 6 with w = [2, -1]. Can a later step receiving only this **score** distinguish those two records?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Dot products and weighted sums case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Multiply matching components then add

POSITION	CONTRIBUTION RULE	WORKED CONTRIBUTION
1 width	2 times width	2 times 4 gives 8
2 height	-1 times height	-1 times 2 gives -2
Add both	8 plus -2	Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compute a new dot product independently

Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Dot products and weighted sums, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

By the end, I can	How I will show it
Calculate a dot product by matching feature positions	Explain the deciding step, show a trace or test, and state one limit.
Explain how a weight changes a score with other inputs fixed	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a weighted score from a probability or trained model	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Dot products and weighted sums without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate a **dot product** by matching feature positions

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Weight**ed sum=6-5=1; this is a **score**, not a probability.

1. Multiply second **input**

2. Add **contribution**s

3. Multiply first **input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Add **contribution**s **Weight**ed sum=6-5=1; this is a **score**, not a probability. A **weight** controls how strongly an **input** contributes to the **score**.

The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase only the right **input** from 5 to6. Which conclusion is supported by the supplied facts?

1. The **score** becomes 6-6=0.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **score** becomes 6-6=0. The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Repair a missing building block

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

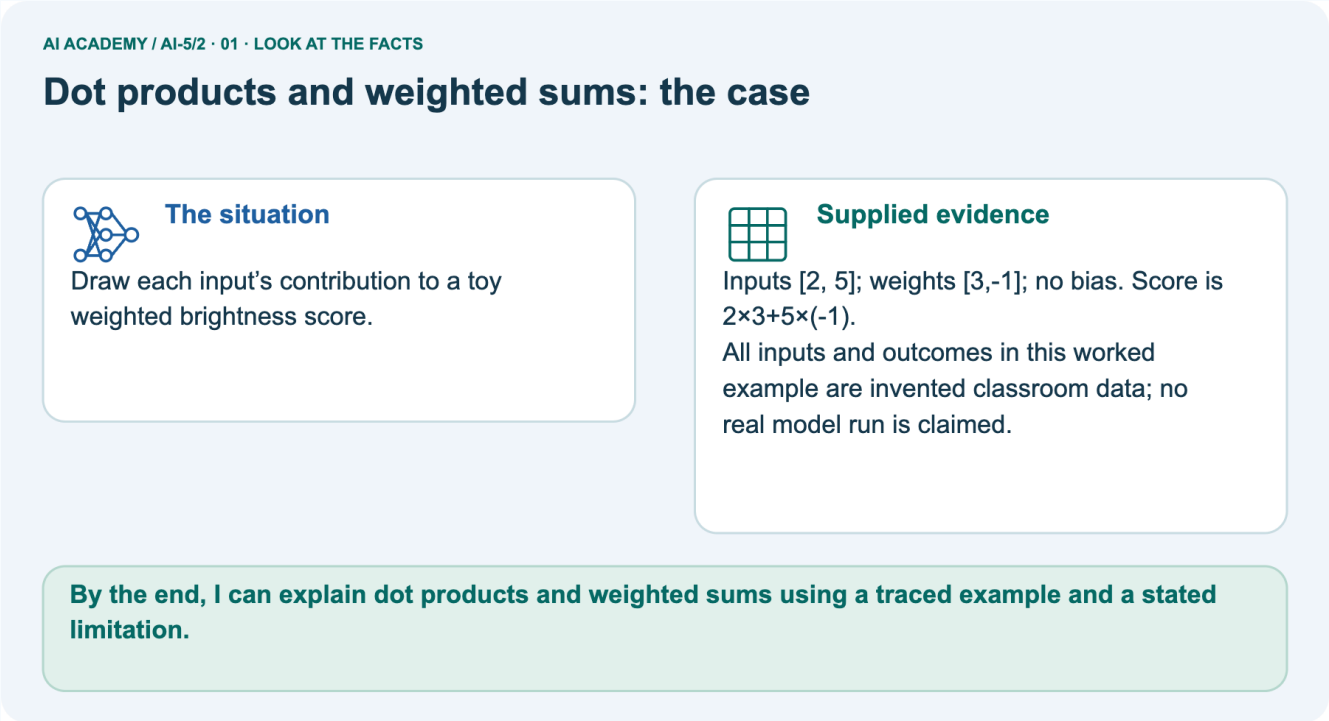
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Dot products and weighted sums graphic using its labels, deciding evidence and relevant limitation.

01. Dot products and weighted sums: the case



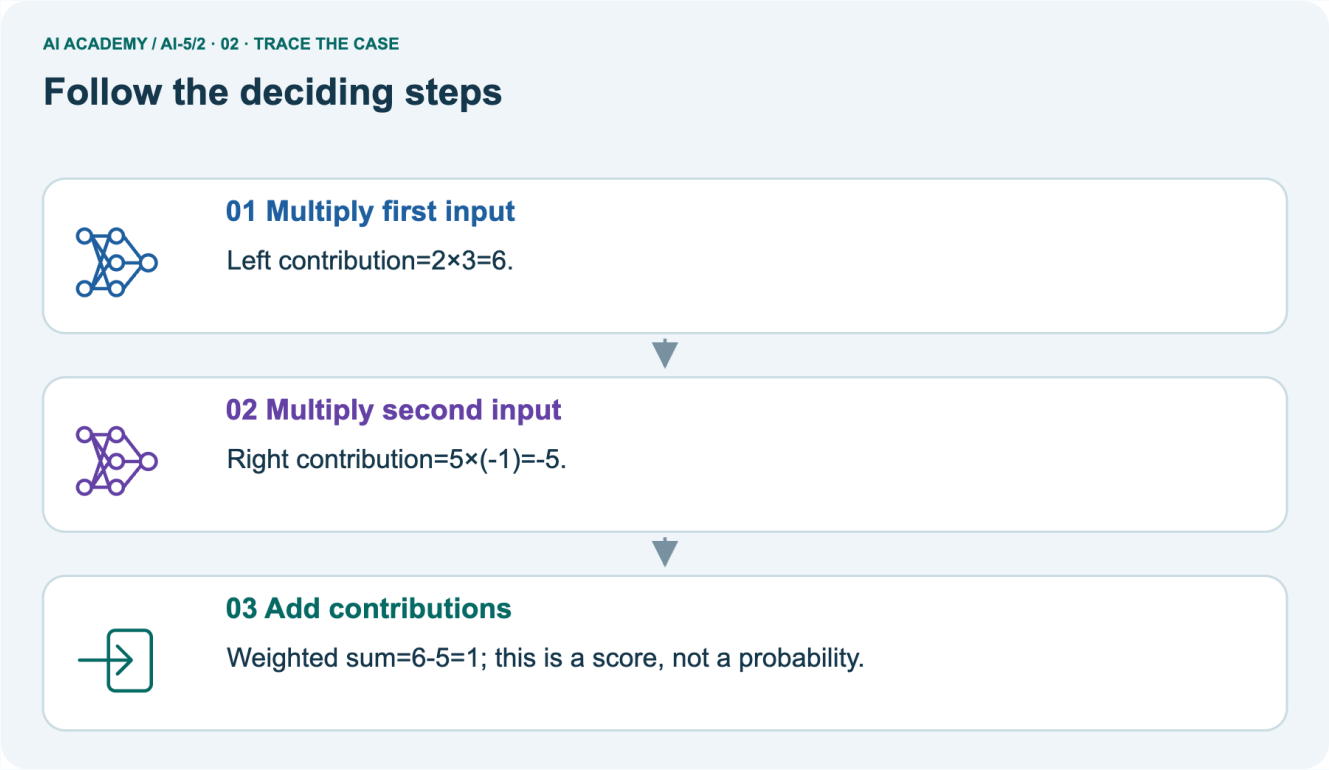
Follow the labelled picture with the matching explanation. Open the vector diagram

Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**. Supplied facts: **Input**s [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

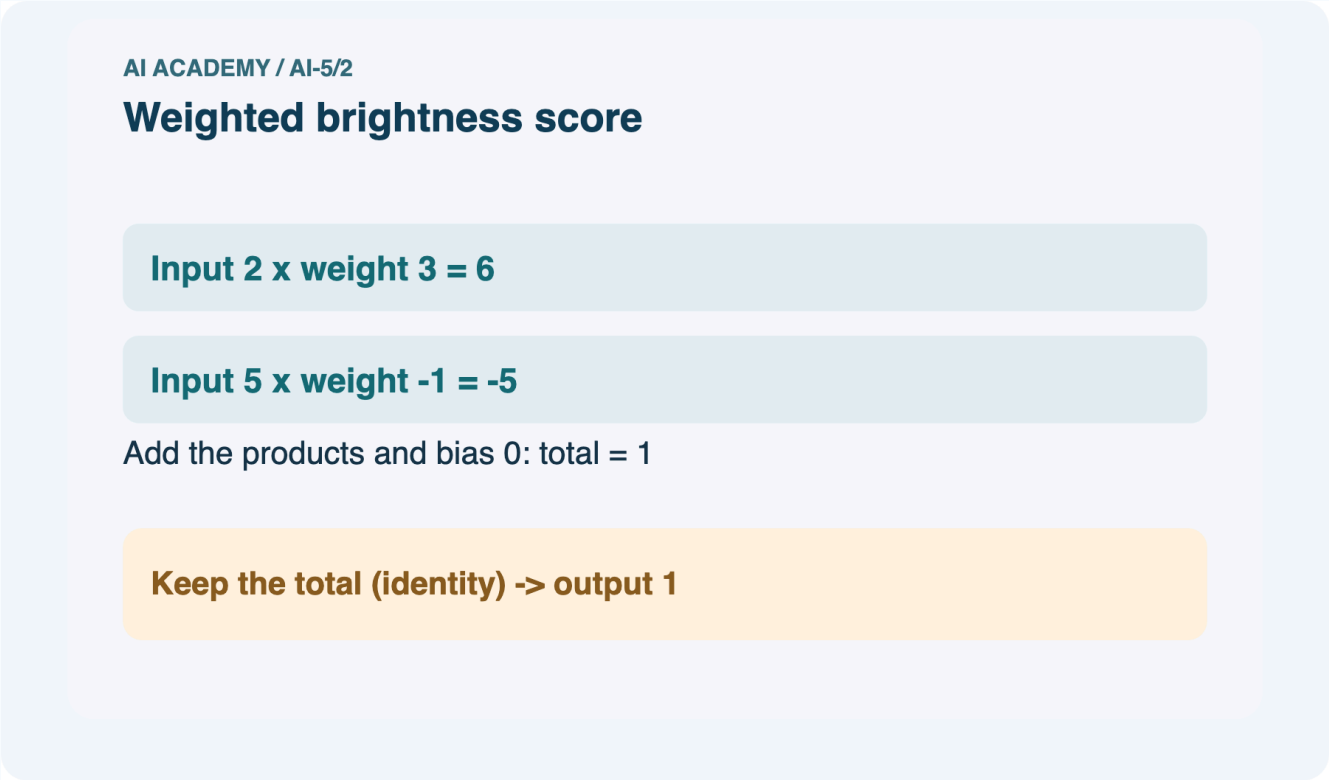


Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply first **input**: Left **contribution**=2×3=6. → Multiply second **input**: Right **contribution**=5×(-1)=-5. → Add **contribution**s: **Weight**ed sum=6-5=1; this is a **score**, not a probability.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Weighted brightness score

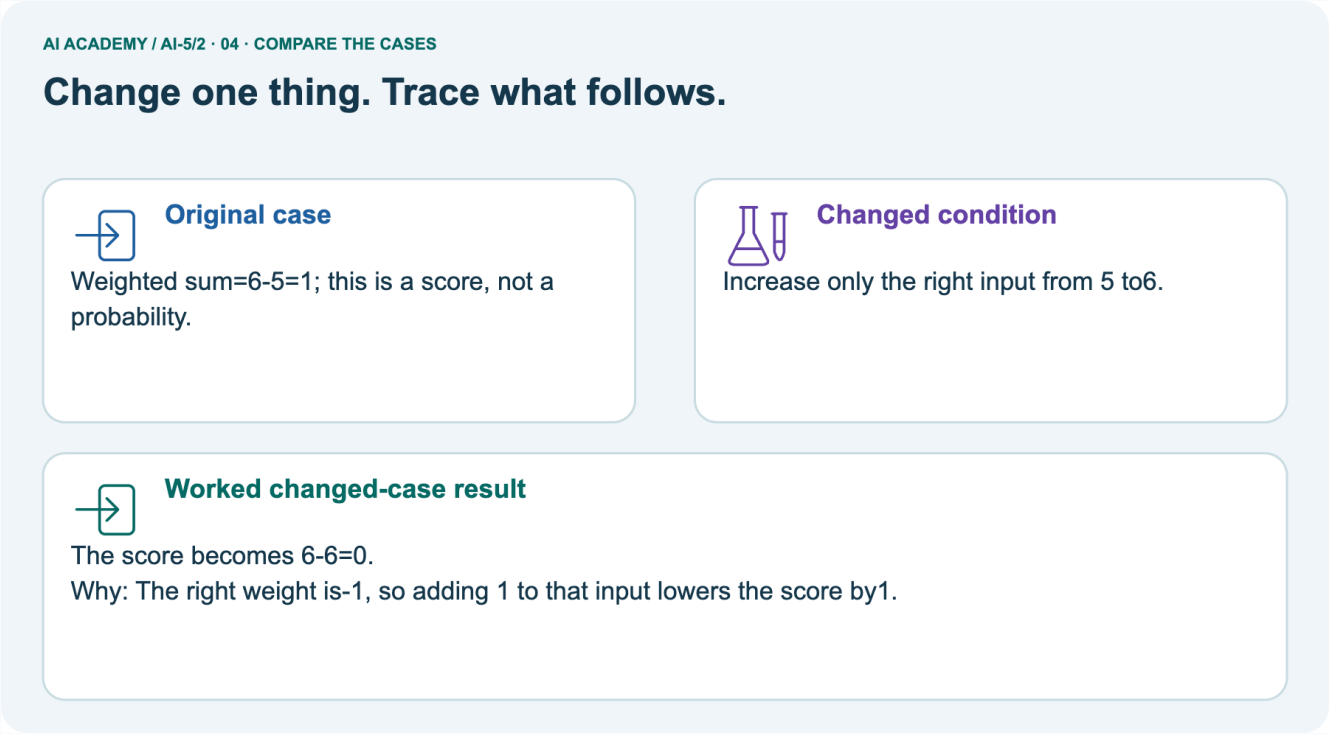


Follow the labelled picture with the matching explanation. Open the vector diagram

Weighted brightness **score**. **Input**s: [2, 5]; **weight**s: [3, -1]; bias: 0; activation: identity; **output**: 1.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



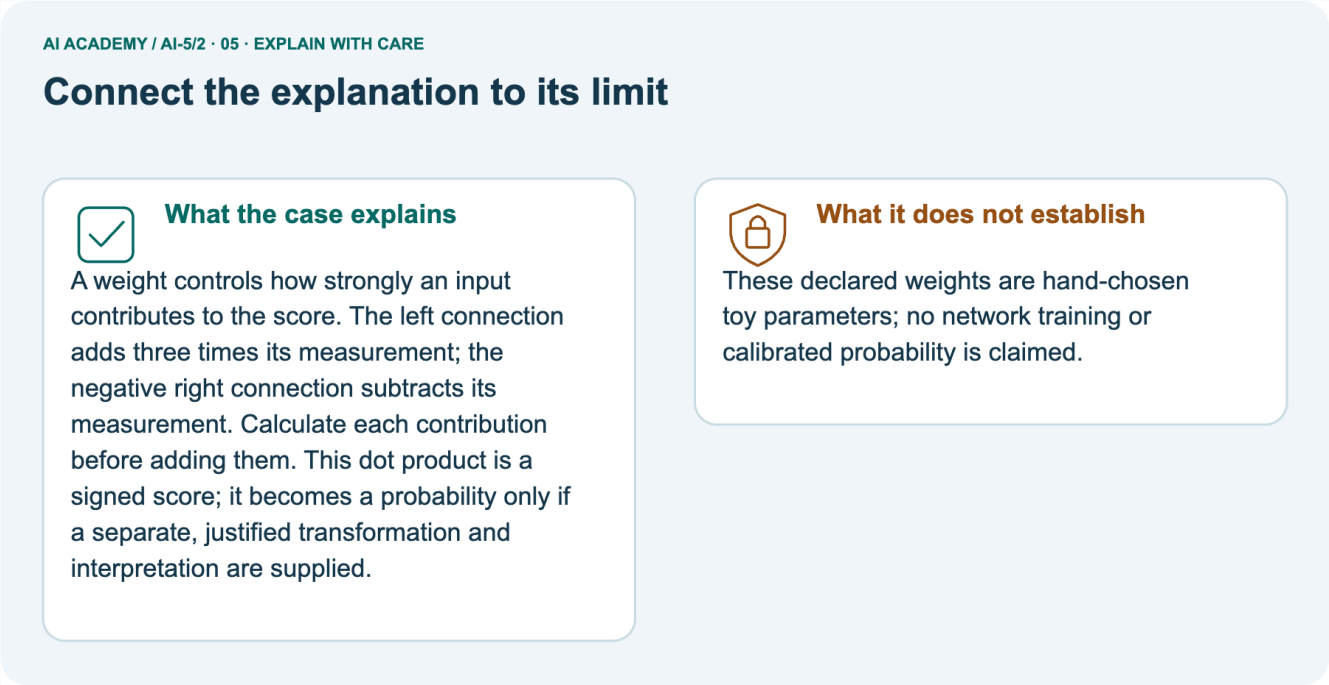
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Weight**ed sum=6-5=1; this is a **score**, not a probability. Changed condition: Increase only the right **input** from 5 to6. Expected result: The **score** becomes 6-6=0.

Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

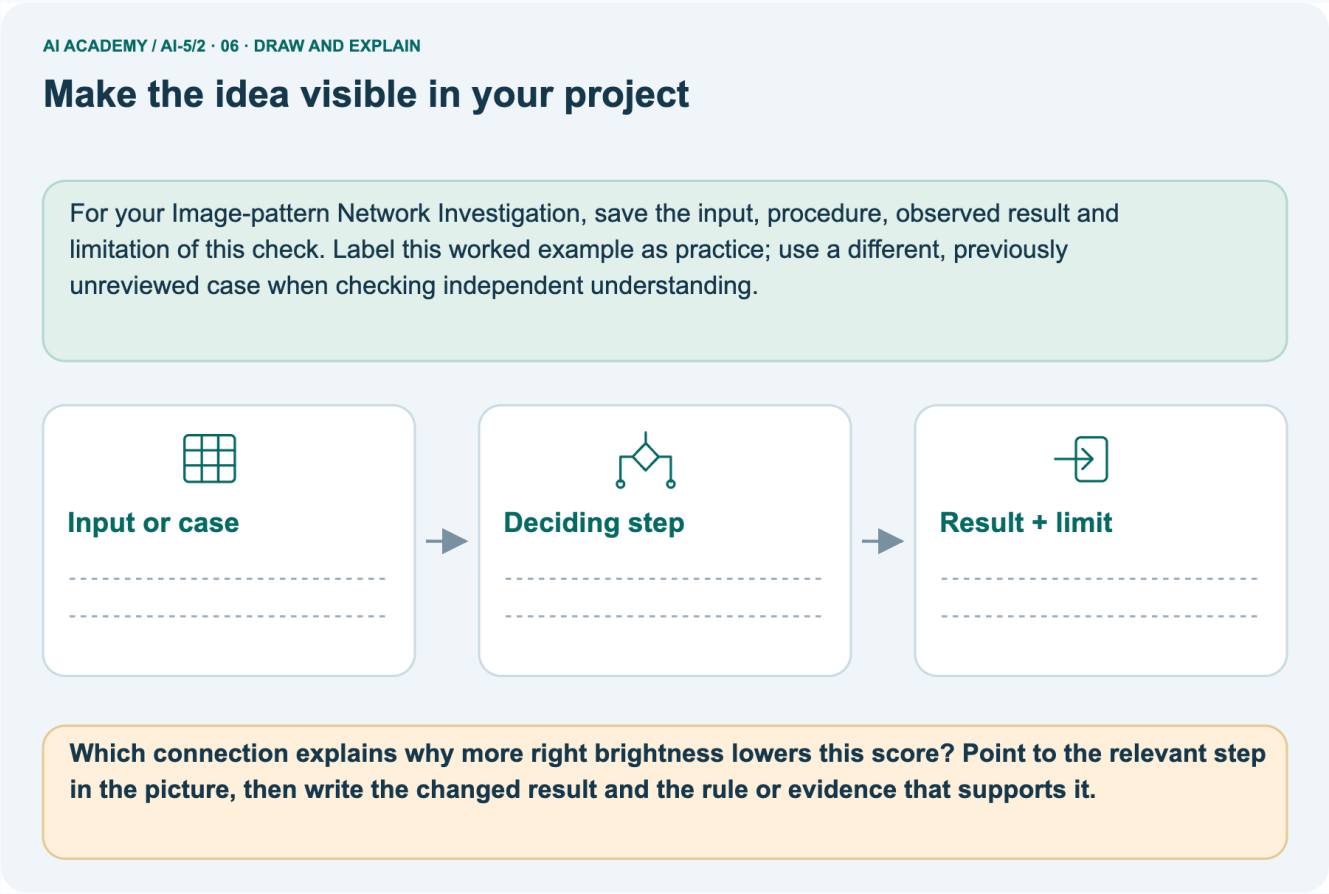
A **weight** controls how strongly an **input** contributes to the **score**. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

Limit: These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.